Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Summary

Instead of providing a classic template to be filled in by authors, we decided that it would be more useful to present with these Instructions for Authors an example that makes it easy to build LaTeX source files designed for the review process of Acta Acustica united with Acustica. The .tex source file of the required format can be downloaded from the Acta Acustica united with Acustica webpage. To ensure the submission process as fast as possible, attention of the authors is paid on the style, design and format of figures that will be submitted (see §2.3): vector graphic format (except for photographs), separate files, size of a width of 75 mm (text should remain readable at this size), lines must be distinguishable in black and white printing (only the online version of the manuscript is in colour). Attention should be drawn to the description of the new paid open access publication option for research articles and paid open access fast track article type (see Sections 3.1 and 6) as an alternative to publishing a paper as a subscription article without a publication fee.

1 Function and scope

Acta Acustica united with Acustica, published in collaboration with the European Acoustics Association (EAA), is an international, peer-reviewed journal on acoustics. It publishes original articles on all subjects in the field of acoustics, including

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The journal reports on original scientific research in acoustics and on engineering applications. The journal considers review papers, scientific papers, technical and applied papers, short communications, letters to the editor, and fast track articles. From time to time, special issues and review articles are also published. For book reviews or doctoral thesis abstracts, please contact the EAA.

2 Manuscripts

2.1 Text processing

The Editors of Acta Acustica united with Acustica recommend that manuscripts should be written in LaTeX [1, 2]. This ensures that the manuscript is easily readable and greatly reduces costs. As a benefit to the author, the paper can be published considerably faster. Authors not familiar with LaTeX can use open source document processing systems based on top of the LaTeX typesetting system (e.g. Lyx, http://www.lyx.org) or may submit their manuscript as Word, WordPerfect, or RTF.

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2.3 Style

2.3.1 Formatting text

Any submitted manuscript should be in a form that allows the referees efficient study. No special formatting is required, but it should be:

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Gambar 1: Recent values of the Impact Factor (IF) of AAA as computed by Thomson-Reuters. Please do not forget that the IF is first, a measure of the size of a discipline.

Gambar 2: (Colour online) Top: total time from first submission to first Editor decision (in days) since 3 years. The values of the most recent months are not significant, and have been zeroed. Both the Editorial Board and the reviewers do their best to provide useful reviews in a reasonably short time! Bottom: monthly number of spontaneous submission (thick, black line); monthly number of rejected papers (thin, grey line).

- in two-column format (width of about 80mm each).
- Pages should be consecutively numbered;
- on each page lines should be numbered;
- the name of the first author should be repeated in the upper right-hand corner of each page.

The sheets of the manuscript should not be bound or stapled. Formulae should be clearly written using standard symbols which are explained at their first appearance. Nomenclatures or lists of symbols will be dropped.

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In order to ensure a readable appearance in the printed version, authors are asked to adhere strictly to some guidelines for the design of figures (see for example Figures 1 and 2):

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For Latex contributions, if the bibliography is done in an external .bib file, authors are asked to upload both the .bib and the .bbl files obtained by a "bibtex" compilation of the main .tex file on their own computer.

The official abbreviation for Acta Acustica united with Acustica is "Acta Acust united Ac".

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New and revised manuscripts shall be submitted online at the EM website, http://www.editorialmanager.com/aaa/ (see also the instruction for authors and reviewers given by Editorial Manager). The following information will need to be provided during submission:

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Scientific Papers contain original material (ideas, models, experiments) not published elsewhere, that contributes substantially to the advance of science in the field of acoustics; they should clearly establish the relation between the work reported on, and the state-of-the-art; scientific papers are typically not longer than 12 pages.

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For accepted scientific papers, technical and applied papers and review papers there is a paid open access publication option upon acceptance of a manuscript (alternatively to publishing a paper as a subscription article without a publication fee) (see Section 6.1).

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The manuscript title should be chosen carefully to reflect the content of the manuscript. No claims for novelty should appear in the title.

A short title should be suggested to appear in the heading of each page of the article.

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The corresponding author should be fully registered in the online manuscript handling system. For other authors that are not registered, entering name and affiliation is sufficient. The submitting author can select a different author to become the corresponding author.

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Manuscripts will be published and indexed under this Field in Acta Acustica united with Acustica.

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Pustaka

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